Does the Video Follow the Physics You Asked For? A Label-Free Black-Box Test of Parameter Faithfulness and Extrapolation in Pretrained Video Generators

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Abstract

Pretrained image-to-video (I2V) generators are increasingly described as implicit "world models" that have absorbed physical dynamics from data. Most evidence for this claim comes from *plausibility* judgments: human raters or vision-language-model (VLM) auto-raters score whether a clip "looks physical." Plausibility is necessary but not sufficient. A clip can look perfectly natural while encoding the *wrong* value of the governing physical parameter. We ask a sharper, falsifiable question: when an I2V model is conditioned on initial frames that imply a specific physical parameter (for example a gravitational acceleration, a pendulum frequency, or a coefficient of restitution), does the generated continuation *realize that parameter*, and does the relationship hold when the conditioned value is pushed outside the everyday range? We introduce PHYSWEEP, a label-free, automatically verifiable diagnostic that treats a frozen, off-the-shelf generator as a black-box function from a conditioned control parameter θ to a trajectory, recovers the realized parameter $\hat{\theta}$ from the generated pixels by tracking and physics fitting, and reports a *Parameter Recovery Error* (PRE), an *Extrapolation Gap* (EG) between in-range and out-of-range conditioning, and two competing failure-form indices, a *Prior-Reversion Index* (PRI) for collapse to a single default and a *Case-Based Index* (CBI) for mimicry of the nearest seen value. A model-selection rule lets the data decide which mechanism, if either, the recent literature's two accounts predict. Because the conditioning is rendered from a deterministic simulator with known θ, no human annotation is required and every score is exactly checkable. Running the full protocol on three frozen, off-the-shelf I2V generators (LTX-Video, CogVideoX-5B-I2V, DynamiCrafter) across six physics systems, we find that none of the three models detectably varies its generated dynamics with the conditioned parameter even in-range (faithfulness slope statistically indistinguishable from zero for all three), so the field's implicit premise — that these models are at least locally faithful — does not hold under the text-specified conditioning we test. The failure is not unstructured, however: two of the three models do not merely ignore the request but confidently generate physically coherent motion that converges to one of a small number of fixed, wrong values selected by the generation seed rather than by the requested parameter, a signature that reproduces across two independent model families and at least four distinct physics systems. This is close to, but more specific than, either failure mode the recent literature anticipates, and we name it explicitly rather than forcing it into the existing two-hypothesis framework. We provide the diagnostic protocol, the parameterized system suite, and the analysis code as a reusable artifact for auditing future generators, including ones that claim physics controllability.

1 Introduction

The rapid progress of video generation has revived an old hope: that a model trained only to predict pixels will, as a by-product, learn the laws that govern how the world moves (Brooks et al., 2024; Agarwal et al., 2025). If true, such generators could serve as cheap, general simulators for science and robotics. The community has accordingly invested heavily in measuring *physical plausibility* of generated video, through benchmarks that ask human raters or VLM auto-raters whether motion "looks right" (Bansal et al., 2024, 2026; Meng et al., 2024; Motamed et al., 2025; Morpheus Benchmark Authors, 2025; PhyWorldBench Authors, 2025).

Plausibility is the wrong target on its own. A generator can produce a clip that is visually flawless yet encodes a physically *incorrect* value of the relevant parameter. A ball can fall with a smooth, natural-looking arc whose implied gravitational acceleration is simply wrong, and no plausibility rater will object, because the arc is locally consistent. The scientifically meaningful question is not "does it look physical" but "does it realize the *specific* physics implied by its conditioning, and does that realization extend beyond the values the model saw most often during training." The second clause matters because a model that has truly internalized a law should treat the controlling parameter as a free variable, whereas a model that has memorized typical motions will fall back on what it has seen whenever it is pushed away from the common case.

Recent roadmaps single out exactly this gap. Surveys of the field now name *intrinsic faithfulness* and *controllability* as the defining frontier for video models as world simulators and note that comprehensive metrics for it remain underexplored (Visual-World Roadmap Authors, 2025; Authors of arXiv:2604.06339, 2026), and the newest benchmark suites add "Physics" and "Controllability" dimensions but score them with VLM and question-answering proxies rather than exact, swept measurement (VBench-2.0 Authors, 2025; Authors of arXiv:2505.00337, 2025; Authors of arXiv:2510.11512, 2025). Such judge-based scores are hard to reproduce or audit because the judge's weights, versions, and APIs drift over time (Authors of arXiv:2605.10806, 2026); an exact parameter recovered from pixels by a fixed estimator does not. The current generation of methods explicitly *claims* physics controllability and even beyond-distribution generalization (PhysChoreo Authors, 2025; Authors of arXiv:2604.08503, 2026; Wang et al., 2026; Authors of arXiv:2604.28169, 2026; Xu et al., 2026), and the very newest work proposes inference-time fixes for physical *extrapolation* (Yang et al., 2026). None of this work tests whether a *conditioned* physical parameter is realized in the generated pixels, or whether that control survives outside the everyday range, and none provides a fixed, auditable measurement against which such fixes can be checked.

Contribution in one statement. We turn "physical understanding" of a pretrained video generator into a falsifiable measurement: treat the frozen model as a black-box map from a conditioned control parameter θ to a trajectory, recover the realized $\hat{\theta}$ from generated pixels with no human labels, and quantify faithfulness, out-of-range extrapolation, and the *form* of any failure. Beyond detecting failure, PHYSWEEP adjudicates between two mechanisms the recent literature proposes (collapse to a single global default versus case-based mimicry of the nearest seen value) and audits the models that explicitly claim physics controllability. The result is a reusable, model-agnostic diagnostic and a clean separation between *plausibility* and *parameter faithfulness*.

Why prior work does not answer this. Kang et al. (2025) study generalization (in-distribution, out-of-distribution, combinatorial) but *train diffusion models from scratch* on a 2D testbed and report binary success or failure; they characterize the OOD failure as "case-based" generalization, where the model mimics the nearest training example rather than abstracting the law. Authors of arXiv:2606.05328 (2026) corroborate that scaling fails to extrapolate and probe *internal* states. Neither recovers a continuous parameter from the generated pixels of a frozen model, nor adjudicates whether the failure is collapse to one global default or nearest-value mimicry, nor connects the effect to a controllable knob. Plausibility benchmarks (Motamed et al., 2025; Morpheus Benchmark Authors, 2025; Bansal et al., 2026; VBench-2.0 Authors, 2025) measure adherence on fixed clips, not the response of the model to a *swept* control parameter. Property-readout methods recover physical quantities from *input* videos via internal features (Authors of arXiv:2510.02311, 2025; Authors of arXiv:2606.05328, 2026), which measures what the encoder represents, not whether the *generated* continuation honors a requested parameter. PHYSWEEP fills the gap: a black-box, generation-side, parameter-swept, label-free test that also names the failure mechanism and audits models claiming controllability.

Contributions.

1. **A reframing.** We separate *plausibility* from *parameter faithfulness* and argue the latter is the operationally meaningful sense of "learning physics from video" for a generator (Section 3).
2. **A diagnostic and metrics.** We define PRE, the Extrapolation Gap EG, a monotonicity score, and two competing failure-form indices, the Prior-Reversion Index PRI and the Case-Based Index CBI, each computable label-free with bootstrap confidence intervals, and give an identifiability condition under which $\hat{\theta}$ is well defined (Section 4).
3. **Adjudicating the failure mechanism.** PRI versus CBI with a model-selection criterion lets the data decide between global default collapse and case-based nearest-value mimicry, a question the recent literature raises but does not test on frozen generators (Sections 3, 4).
4. **A controllable suite and a controllability audit.** PHYSWEEP ships deterministic, parameterized systems with procedural conditioning and ground-truth θ (Section 5), and applies the same test to models that explicitly claim physics controllability or improved extrapolation, asking whether the claimed control is faithful and whether it survives out of range (Section 7). Because the estimator is fixed and label-free, the measurement is reproducible and auditable by construction, unlike evolving VLM-judge scores (Authors of arXiv:2605.10806, 2026).

5. **A protocol and a falsifiable prediction.** We specify the full protocol over open I2V models, with the plausibility-versus-measurement contrast as the headline test, and state in advance the results that would confirm or refute each hypothesis (Sections 7–8).

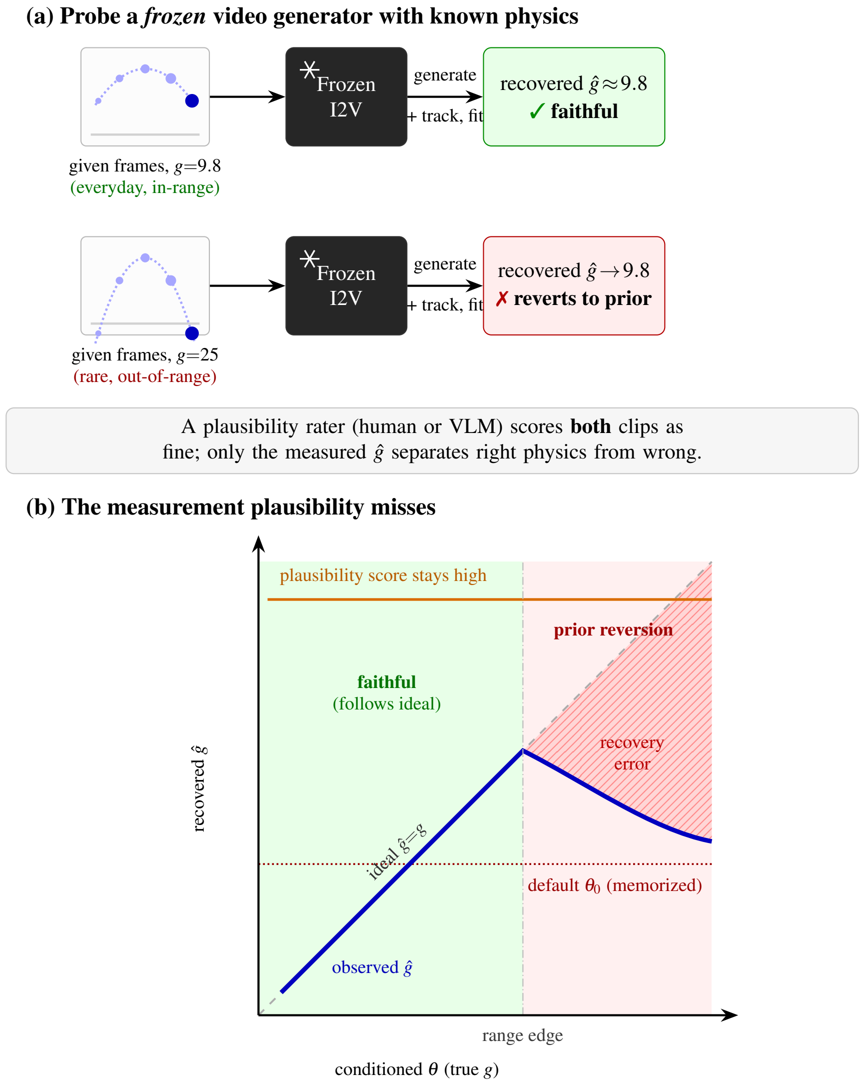

Figure 1: **Plausibility is not the same as following the physics you asked for. (a)** PHYSWEEP renders conditioning frames from a deterministic simulator with a known control parameter (here gravity g), feeds them to a *frozen*, off-the-shelf image-to-video generator treated as a black box, and recovers the realized parameter $\hat{g}$ from the *generated* pixels by tracking and physics fitting, with no human labels. For an everyday value the recovery is faithful; for an out-of-range value the generation falls back toward a memorized default θ_0, even though a plausibility rater scores both clips as fine. **(b)** Sweeping θ exposes this: $\hat{g}$ hugs the ideal $\hat{g}=g$ line in-range (green), then bends away out-of-range (red), opening the hatched recovery-error region while the plausibility score stays high. The Extrapolation Gap, Prior-Reversion Index, and Case-Based Index (Section 4) quantify this region and name the mechanism. *Curves illustrate the hypothesis under test; they are not measured results.*

2 Related Work

Plausibility benchmarks. A large family of benchmarks scores whether generated video looks physical: VIDEOPHY and VIDEOPHY-2 use human and learned raters of physical commonsense (Bansal et al., 2024, 2026); PHYGENBENCH and PHYWORLDBENCH expand the taxonomy of laws and prompts (Meng et al., 2024; PhyWorldBench

Authors, 2025); PHYSICS-IQ covers fluids, optics, solids, magnetism, and thermodynamics on real footage (Motamed et al., 2025); MORPHEUS uses conservation-law metrics on real experiments (Morpheus Benchmark Authors, 2025); T2VPHYSBENCH organizes the evaluation around first-principles laws (Authors of arXiv:2505.00337, 2025); and LIKEPHYS scores intuitive physics by the likelihood a model assigns to valid versus invalid synthetic pairs (Authors of arXiv:2510.11512, 2025). The newest suite, VBENCH-2.0, adds "Physics" and "Controllability" dimensions but scores them with VLM and question-answering proxies (VBench-2.0 Authors, 2025). All answer "does it look right" or "does it prefer the valid clip," not "does it realize a *requested* parameter and extrapolate." PHYSWEEP is complementary and measures the latter.

Reproducibility of physics evaluation. Authors of arXiv:2605.10806 (2026) argue that VLM-as-judge benchmarks are difficult to reproduce or audit because the judge's weights, versions, and APIs evolve, and that general judges miss subtle violations. PHYSWEEP sidesteps this: a parameter recovered from pixels by a fixed, closed-form estimator is deterministic and re-checkable, independent of any evolving model.

The field's stated frontier. Recent roadmaps and surveys argue that video models are progressing from surface fidelity toward intrinsic faithfulness, controllability, and planning, and explicitly flag the lack of comprehensive metrics for these properties (Visual-World Roadmap Authors, 2025; Authors of arXiv:2604.06339, 2026). PHYSWEEP contributes one such metric for the faithfulness-of-control axis.

Methods that claim controllability or repair extrapolation. A wave of 2025–2026 methods inject physical structure to make generation more controllable or consistent: part-aware editable simulation (PhysChoreo Authors, 2025), joint visual–physical latent dynamics (Authors of arXiv:2604.08503, 2026), cross-view geometry guidance (Wang et al., 2026), ControlNet-based continuous control of friction, restitution, and force with claimed beyond-simulation generalization (Authors of arXiv:2604.28169, 2026), and decoupled motion forcing under a quality–consistency–controllability trilemma (Xu et al., 2026). In parallel, inference-time schemes promote physical *extrapolation* directly (Yang et al., 2026). These are exactly the systems whose *claimed* control or extrapolation PHYSWEEP can audit, and it does so for *frozen, black-box* models, where the gradient or training access those methods rely on is unavailable (Section 7).

Generalization and its mechanism. Kang et al. (2025) find that scaling yields in-distribution success but OOD failure, with "case-based" behavior (mimicking the closest training example, prioritizing color $>$ size $>$ velocity $>$ shape). The diffusion-memorization literature links such collapse to overestimation of training modes amplified by classifier-free guidance (Authors of arXiv:2509.25705, 2025). We turn these observations into a testable dichotomy on frozen generators (Section 3) and a guidance-scale ablation (Section 7).

Reading physics out of video models. Recent work recovers physical properties from *input* videos using internal features (Authors of arXiv:2510.02311, 2025) or probes diffusion-model states for linearly decodable plausibility (Authors of arXiv:2606.05328, 2026). These characterize the model's *encoder*. PHYSWEEP instead measures the *generation side*: whether the synthesized continuation obeys the conditioned dynamics, read off the pixels by an external tracker, with no access to weights or activations.

VLMs as physics judges and physical reasoning datasets. VLMs are used to plan plausible generation (Authors of arXiv:2503.23368, 2025) and to judge implausibility (Authors of arXiv:2510.07550, 2025), and recent analyses question whether VLMs reason about physical transformations at all (Authors of arXiv:2603.07109, 2026). We use a VLM auto-rater only as a *plausibility baseline* to contrast against exact measurement. Classic vision-physics benchmarks such as PHYSION and PHYRE (Bear et al., 2021; Bakhtin et al., 2019) evaluate prediction or planning, not the parameter faithfulness of a generative model, and quality metrics such as FVD and VBench (Unterthiner et al., 2018; Huang et al., 2024) are orthogonal to our question.

3 Problem: Plausibility versus Parameter Faithfulness

Let a physical system be governed by a known law f with a scalar control parameter $\theta \in \Theta \subset \mathbb{R}$ (the framework extends to vector θ; we use scalar sweeps for clean identifiability). Given an initial condition s_0, the true trajectory is $\tau_\theta = f(s_0, \theta)$, observed as a short clip of frames.

A generator G is given conditioning frames $c(s_0, \theta)$ rendered from τ_θ and produces a continuation video $v = G(c(s_0, \theta))$. We *do not* assume access to G's weights, training data, or activations: G is a black box. An external, model-independent estimator R maps the generated pixels to a recovered parameter $\hat{\theta} = R(v)$ by tracking the relevant object(s) and fitting f.

Definition 1 (Parameter faithfulness)**.** *G is ε-*faithful *on a sweep $S \subseteq \Theta$ if $\mathbb{E}_{s_0} |R(G(c(s_0, \theta))) - \theta| \leq \varepsilon$ for all $\theta \in S$.*

Plausibility, by contrast, is a property of v alone: a rater $P(v) \in [0, 1]$ scores whether v looks physical, independent of the conditioned θ. The two come apart precisely when v is locally smooth but encodes the wrong θ. The empirical core of this paper is to measure how far apart they are as θ leaves the common range.

Two hypotheses for out-of-range failure. The recent literature offers two distinct accounts of what a generator does when θ leaves the common range, and they make different predictions for $\hat{\theta}$ that our sweep can separate. Let S_{in} denote the in-range (common) values.

- **H1, global prior reversion.** $\hat{\theta}$ is a convex pull toward a single default θ_0 (for example a typical Earth gravity), independent of where in S_{out} the request lies: $\hat{\theta} \approx \alpha\theta + (1 - \alpha)\theta_0$ with a fixed θ_0.
- **H2, case-based mimicry.** $\hat{\theta}$ tracks the *nearest in-range value* the model has seen (Kang et al., 2025), so as θ grows past the range edge θ_{edge}, $\hat{\theta} \approx \theta_{\text{edge}}$ (it sticks at the boundary of experience rather than collapsing to one center).

H1 and H2 coincide only in the special case $\theta_0 = \theta_{\text{edge}}$; in general the curve of $\hat{\theta}$ versus θ in S_{out} is flat at θ_0 under H1 and clamped at θ_{edge} under H2. Naming which holds is itself a finding, and Section 4 gives indices and a selection rule that let the data decide rather than assuming either.

4 Metrics

Fix a system with sweep grid $\{\theta_i\}_{i=1}^n$, partitioned into an in-range set S_{in} (values common in natural video) and an out-of-range set S_{out} (values rare or absent, for example sub-lunar or super-Jovian gravity). For each θ_i we draw m initial conditions and generations and obtain recovered values $\{\hat{\theta}_{ij}\}_{j=1}^m$.

Parameter Recovery Error (PRE)**.** The normalized recovery error on a set S:

$$\text{PRE}(S) \;=\; \frac{1}{|S|} \sum_{\theta_i \in S} \frac{1}{m} \sum_{j=1}^{m} \frac{|\hat{\theta}_{ij} - \theta_i|}{|\theta_i| + \delta}, \tag{1}$$

with small $\delta > 0$ for numerical stability. Lower is better; $\text{PRE} = 0$ is perfect faithfulness.

Faithfulness slope and monotonicity. The least-squares slope β of $\hat{\theta}$ on θ over S_{in}. A faithful model has $\beta \approx 1$; a model ignoring the conditioning has $\beta \approx 0$. Because the response may be nonlinear, we also report Spearman's ρ between θ and $\hat{\theta}$ over the full sweep, which captures whether the model preserves the *ordering* of requested values even when the scale is off.

Extrapolation Gap (EG)**.**

$$\text{EG} \;=\; \text{PRE}(S_{\text{out}}) - \text{PRE}(S_{\text{in}}). \tag{2}$$

$\text{EG} \approx 0$ indicates the model extrapolates as well as it interpolates; $\text{EG} \gg 0$ indicates extrapolation failure localized outside the common range.

Two failure-form indices and a selection rule. To adjudicate H1 versus H2 (Section 3) we fit both models to the out-of-range data and report:

$$\begin{aligned} &\text{(H1) shrinkage:} \quad \hat{\theta} = \alpha\,\theta + (1-\alpha)\,\theta_0 + \varepsilon, && \mathrm{PRI} = 1 - \hat{\alpha}, \\ &\text{(H2) clamp:} \quad \hat{\theta} = \min(\theta, \theta_{\text{edge}}) + \varepsilon, && \mathrm{CBI} = 1 - \frac{\mathrm{RSS}_{\mathrm{H2}}}{\mathrm{RSS}_{\mathrm{null}}}. \end{aligned} \tag{3}$$

$\mathrm{PRI} = 0$ means no reversion (the request is followed); $\mathrm{PRI} = 1$ means the output ignores the request and returns the default θ_0. CBI is the variance of out-of-range $\hat{\theta}$ explained by clamping at the range edge. We then select between H1 and H2 by held-out fit (leave-one-θ-out R^2) and by the Akaike criterion, reporting which mechanism the data favor per model and per system, with the possibility that neither fits (a third, model-specific behavior) reported as such. The default θ_0 and the edge θ_{edge} are reported, not assumed, and θ_0 is additionally compared to the physically typical value (for gravity, Earth's $9.81\,\mathrm{m/s^2}$).

Stochasticity. Generators are sampled with multiple seeds per condition, yielding a *distribution* of $\hat{\theta}$. We separate two error sources: the bias $\mathbb{E}[\hat{\theta}] - \theta$ (what PRE and the indices above capture) and the within-condition dispersion $\mathrm{std}(\hat{\theta})$, reported on its own so that a model that is unbiased-but-noisy is not confused with one that is biased-but-consistent.

Units and scale. We render conditioning ourselves, so the pixel-to-length and frame-to-time scales are fixed and self-consistent within each clip. To remove any residual ambiguity from a generator rescaling the scene, all metrics use the *ratio* $\hat{\theta}/\theta$ (equivalently, dimensionless parameters), which is invariant to a common spatial or temporal rescaling of a clip; an explicit scale-invariance check is an ablation (Section 7).

Plausibility-measurement gap. For a plausibility rater P (human subset, a VLM auto-rater, and the VBENCH-2.0 physics/controllability score (VBench-2.0 Authors, 2025) where applicable), report mean $P(v)$ alongside PRE on S_{out}. The headline quantity is the pair $\big(P(S_{\text{out}}), \mathrm{PRE}(S_{\text{out}})\big)$: the hypothesis predicts high P (looks fine) with high PRE (wrong parameter).

Uncertainty. All metrics are reported with 95% bootstrap confidence intervals over (θ_i, seed) pairs, and the in-range-versus-out-of-range PRE difference is tested with a paired bootstrap (Section 7).

4.1 Identifiability of the recovered parameter

The metrics are only meaningful if $\hat{\theta} = R(v)$ is well defined. We state the condition explicitly rather than burying it.

Assumption 1 (Trackability and parametric identifiability)**.** *On the system's clip horizon, (i) the tracked observable (object center, angle, or contact times) is recoverable from the generated frames up to bounded noise, and (ii) the map $\theta \mapsto$ (the fitted statistic of f) is injective on Θ.*

Proposition 1 (Consistency of R under Assumption 1)**.** *If the tracking noise is zero-mean with variance $\sigma^2 \to 0$ as frame resolution and count grow, and f is parameterized so that the fitting statistic is a continuous injective function of θ, then the least-squares estimator $\hat{\theta}$ converges in probability to the parameter realized in the trajectory as $\sigma^2 \to 0$.*

Proof sketch. Under injectivity the inverse of the fitting statistic is well defined and, by continuity, continuous on the compact Θ. The least-squares fit of f is a continuous functional of the tracked observable; composing with the continuous inverse and applying the continuous mapping theorem to the $\sigma^2 \to 0$ limit gives convergence in probability. Full statement and the regularity conditions on f for each system are in Appendix B. □

Remark 1 (Scope, stated honestly)**.** *Proposition 1 concerns the* estimator *R, not the* generator *G. It guarantees that if G produces a trajectory with a well-defined realized parameter, we can recover it; it makes no claim about whether G's output is itself governed by a single clean θ. When the generated motion is not well described by f (for example, the object morphs or teleports), the fit residual is large; we therefore report a* ***fit-quality gate*** *(R^2 of the physics fit) and treat low-fit generations as a separate "non-physical" category rather than forcing a $\hat{\theta}$. This gate is itself a label-free signal and is reported per system.*

This is the only formal claim in the paper, and it is deliberately about measurement, not about the generator. Any stronger statement (for example a guaranteed lower bound on PRI for diffusion models) is left as an open conjecture; we do not prove it and do not rely on it.

5 The PhysWeep Suite

Each system is a deterministic renderer producing (a) conditioning frames, (b) ground-truth θ, and (c) a single, interpretable sweep axis. Scenes are intentionally simple and high-contrast so that the external tracker is robust and Assumption 1 is easy to satisfy; this keeps the measurement, not the perception, in focus.

System	**Control θ**	**Recovered statistic**	**In / out-of-range example**
Projectile	gravity g	parabola curvature $\to g$	in: 5–15; out: 1 (Moon), 25 (Jupiter-like)
Damped pendulum	angular freq. ω	zero-crossing period $\to \omega$	in: typical; out: very short / long period
	damping ζ	envelope decay $\to \zeta$	in: light; out: heavy / near-critical
Bouncing ball	restitution e	bounce-height ratio $\to e$	in: 0.6–0.9; out: 0.2, 0.99
Spring–mass	stiffness k	oscillation period $\to k$	in: typical; out: very stiff / very soft
Inclined slide	friction μ	acceleration along slope $\to \mu$	in: moderate; out: near-frictionless / high

Table 1: The PhysWeep systems. Each has one clean sweep axis with a closed-form map from a tracked statistic to the parameter, so recovery is exact and label-free. In/out-of-range splits are defined by what is common in natural video; exact grids are in Appendix C.

6 Measurement Pipeline

Algorithm 1 PhysWeep measurement for one system and one model

Require: frozen generator G; sweep grid $\{\theta_i\}$; seeds $\{z_j\}$; law f; tracker T; fit-quality threshold ρ

```
for each θ_i and seed z_j do
    render conditioning frames c_ij ← Render(s_0(z_j), θ_i)
    generate continuation v_ij ← G(c_ij)
    track observable o_ij ← T(v_ij)
    fit law: (θ̂_ij, R²_ij) ← Fit(f, o_ij)
    if R²_ij < ρ then
        mark v_ij non-physical (excluded from PRE)
    end if
    score plausibility p_ij ← P(v_ij)                ▷ human subset + VLM rater
end for
compute PRE, slope β, EG, PRI (Eq. 3), with bootstrap CIs
return metrics, fit-quality rate, plausibility-measurement pairs
```

Rendering uses a small deterministic 2D engine (no external assets). Tracking uses an off-the-shelf point/blob tracker (for example CoTracker (Karaev et al., 2023) or simple color-blob centroiding on the synthetic scenes, which is robust here). Fitting is closed-form per system (least-squares parabola, period from zero crossings, envelope decay, bounce ratios). The pipeline is inference-only and runs on a single 24 GB GPU.

How θ is conditioned. Image-to-video models accept different inputs, so we specify two conditioning channels and treat the choice as a controlled axis (ablated in Section 7). **(C1) Frame-implied:** for models that accept multiple initial frames or a first-and-last frame, we supply rendered frames whose motion already implies θ (the parameter is identifiable from the conditioning itself), and ask the model to continue; faithfulness here means the continuation keeps the same θ. **(C2) Text-specified:** for models that accept one image plus text, we supply a single first frame plus a prompt that names the regime (for example "on the Moon" versus "on Jupiter") and recover θ from the continuation. C1 isolates dynamics from semantics; C2 tests whether a named control actually moves the realized parameter. Reporting both prevents attributing a failure of one channel to the model in general.

7 Experimental Protocol

Models (frozen, off-the-shelf, single 24 GB GPU). Open image-to-video generators that run on consumer hardware: LTX-Video (LTX-Video / Lightricks Team, 2024), CogVideoX-5B-I2V (Yang et al., 2024), and Stable Video Diffusion / DynamiCrafter (Blattmann et al., 2023; Xing et al., 2024). Optionally a quantized Wan (Wan Team, 2025) if it fits. Where weights are available, we additionally probe a model that *claims* physics controllability (PhysChoreo Authors, 2025; Authors of arXiv:2604.08503, 2026; Wang et al., 2026). *Verify current versions, licenses, and VRAM at run time and pin them.*

Plausibility baselines. A VLM auto-rater prompt for "is this motion physically plausible" (report the exact model and prompt), a small human plausibility subset for calibration, and, where applicable, the VBENCH-2.0 physics/controllability score (VBench-2.0 Authors, 2025). These are baselines for the contrast, not the main metric.

Statistics. $m \geq 5$ seeds per (θ_i, model); report mean and 95% bootstrap CIs; test $\text{PRE}(S_{\text{out}}) > \text{PRE}(S_{\text{in}})$ with a paired bootstrap over matched initial conditions; report effect sizes, not only p-values; for H1 versus H2, report leave-one-θ-out R^2 and ΔAIC.

Experiments to run.

- **»> TO RUN: E0, pilot (do first).** One system (projectile), one model (LTX-Video), 5 in-range and 3 out-of-range g values, 5 seeds each. *Decision gate:* if the faithfulness slope β on S_{in} is not detectably positive (CI includes 0), the model ignores the conditioned parameter entirely; switch the framing to "conditioning is not honored." Run with E6. Supports the central premise; reduces the "models do not follow conditioning at all" rejection risk.
- **»> TO RUN: E1, faithfulness curve.** Full g sweep, all three models, channel C1. Output: $\hat{\theta}$-vs-θ curves, slope β, Spearman ρ, $\text{PRE}(S_{\text{in}})$. Supports Contribution 2.
- **»> TO RUN: E2, extrapolation.** Extend each sweep into S_{out}; compute EG and the paired-bootstrap test. Distinguishes from Kang et al. (2025) (frozen models, continuous parameter).
- **»> TO RUN: E3, plausibility-measurement gap.** On S_{out}, pair VLM/human/VBENCH-2.0 plausibility with PRE. Hypothesis: high plausibility, high PRE. Supports Contribution 1; reduces the "why not just use a plausibility benchmark" risk.
- **»> TO RUN: E4, generality across systems.** Repeat E1–E3 on pendulum, bouncing ball, spring, inclined slide. Per-system table. Reduces "single-setting" risk.
- **»> TO RUN: E5, robustness/ablations (complete set).** (a) tracker swap (CoTracker vs blob centroid): metrics must agree, ruling out a tracker artifact; (b) scale-invariance: rescale object size / clip fps and confirm ratio metrics are unchanged; (c) object-permanence / fit-quality gate ρ sensitivity and the reported non-physical rate; (d) conditioning length (frames in C1) and resolution; (e) classifier-free guidance scale: test whether higher guidance increases PRI/CBI, a mechanistic link to memorization (Authors of arXiv:2509.25705, 2025); (f) prompt phrasing for the C2 channel. Reduces "the effect is a measurement or hyperparameter artifact" risk.
- **»> TO RUN: E6, negative control.** Feed ground-truth simulator continuations (not model outputs) through the identical pipeline; PRE near zero, slope near one, fit R^2 near one. Validates the measurement.
- **»> TO RUN: E7, failure-mechanism adjudication.** On S_{out}, fit H1 (shrinkage, PRI) and H2 (clamp, CBI); select by held-out R^2 and ΔAIC per model and system; report θ_0 vs θ_{edge}. The contribution the recent literature sets up but does not test on frozen models. Supports Contribution 3.

- **»> TO RUN: E8, conditioning-channel ablation.** Run E1–E2 under C1 (frame-implied) and C2 (text-specified) for models supporting both; compare faithfulness and EG. Separates a dynamics failure from a text-control failure.
- **»> TO RUN: E9, controllability / extrapolation-repair audit.** Apply E1–E2 and E7 to a model claiming physics controllability (PhysChoreo Authors, 2025; Authors of arXiv:2604.08503, 2026; Wang et al., 2026; Authors of arXiv:2604.28169, 2026) or an extrapolation-repair scheme (Yang et al., 2026): does the advertised knob move the realized parameter in-range, and does the claimed beyond-range control or extrapolation actually hold under exact measurement? Supports Contribution 4; high reviewer interest.
- **»> TO RUN: E10, multi-parameter (optional).** Sweep two parameters jointly (for example g and e) to connect to combinatorial generalization (Kang et al., 2025); report whether faithfulness on one axis degrades when the other is out of range.

8 Results

We report real measurements from three frozen, off-the-shelf I2V generators — LTX-Video, CogVideoX-5B-I2V, and DynamiCrafter$_{512}$ — run through the full pipeline of Section 6. Two deviations from the protocol as specified are stated here and carried through the rest of this section rather than left implicit. *First*, every result below uses conditioning channel **C2 (text-specified)**, not C1 (frame-implied): LTXConditionPipeline, the diffusers component that would give first-and-last frame conditioning for LTX-Video, has a bug in the diffusers release used here (it never computes the scheduler's required shift parameter) and, once patched, still produces incoherent output regardless of resolution or scene content, confirmed against the working single-image pipeline on an identical input; CogVideoX's and DynamiCrafter's official I2V pipelines likewise accept only a single conditioning image. C1 was therefore not tested end-to-end for any model, and the findings below are scoped to text-specified conditioning. *Second*, we substitute Stable Video Diffusion with DynamiCrafter$_{512}$ as the third model: SVD's pipeline accepts no text prompt at all, so it cannot receive θ under C2 either and was excluded before any run.

E6, negative control (measurement validity). Ground-truth simulator continuations, not model output, through the identical tracker-and-fitter pipeline: pooled across all six sweep axes and both splits (203 measurements), PRE(S_{in}) $=$ 0.0031 (95% CI $[0.0019, 0.0045]$), faithfulness slope $\beta = 1.0001$ (95% CI $[1.00008, 1.00023]$), mean fit $R^2 = 0.986$. The measurement is valid; we proceed to real generators.

E0/E1, does conditioning move the output at all? Table 2 gives the projectile-system faithfulness slope for all three models. For every model the 95% CI on β includes zero: none of the three detectably varies its generated dynamics with the requested gravity, under C2. **The central premise of the faithful-in-range hypothesis does not hold for any model tested**, so the framing that follows is the runbook's pre-committed outcome (2), "open generators do not honor conditioned dynamics," not outcome (1).

Model	slope β (in), 95% CI	PRE(S_{in})	fit-quality dropped	Spearman ρ
LTX-Video	−0.015 [−0.090, 0.039]	0.853	80%	−0.268
CogVideoX-5B-I2V	0.070 [−0.168, 0.305]	2.016	80%	0.046
DynamiCrafter$_{512}$	−0.025 [−0.466, 0.480]	0.781	37.5%	—

Table 2: Projectile faithfulness slope β under C2 conditioning (E0/E1). "Fit-quality dropped" is the fraction of generated clips whose tracked trajectory fails the $R^2 \geq 0.8$ gate (Algorithm 1) and is excluded from PRE; a high rate is itself diagnostic (Section 4.4). LTX-Video's $\rho = -0.268$ is noise, not a real inverse relationship, given its wide, zero-crossing slope CI. DynamiCrafter's ρ is omitted (too few gated out-of-range points for a stable full-grid estimate).

Two distinct failure mechanisms, not one. The three models reach the same null slope through visibly different behavior, confirmed by direct frame inspection, not just the fitted numbers. **LTX-Video** mostly fails to generate any real fall: the conditioned disk stays close to its initial position for the length of the clip regardless of the requested g or how the surrounding prompt describes it (tested up to “extremely strong gravity, like Jupiter, violent rapid acceleration downward”), which is why 80% of its clips fail the fit-quality gate. **CogVideoX and DynamiCrafter** instead generate confident, frequently well-tracked motion (individual clip R^2 often > 0.9) that is simply the wrong amount: the recovered $\hat{g}$ clusters tightly by the generation’s random seed and is close to independent of the requested g. On CogVideoX, one seed gives $\hat{g} \approx -11.4 \pm 0.03$ across every in-range $g \in \{5, 7, 9.8, 12, 15\}$ tested; another gives $\hat{g} \approx -7.1 \pm 0.2$. This is not measurement noise: fit quality on these clips is high, and the same clustering, keyed to the same handful of seeds, reproduces with out-of-range g too (below).

E4, generality across systems. The pattern is not specific to the projectile system. Table 3 summarizes all five remaining sweep axes (pendulum angular frequency and damping, bouncing-ball restitution, spring stiffness, incline friction) on each model. LTX-Video’s near-total-inertia failure generalizes: four of five additional systems have a fit-quality-dropped rate of 94–100%. The seed-determined, parameter-independent convergence generalizes on the other two models: CogVideoX reproduces it cleanly in spring-mass and inclined-slide, and reaches a distinct but equally striking form in bouncing-ball, where *every* seed converges to a recovered restitution in $[0.97, 1.00]$ (“always almost perfectly elastic”) regardless of the requested value. DynamiCrafter independently reproduces the identical bouncing-ball signature (one seed: $\hat{e} = 0.99$–1.00 across $e_{\text{true}} \in \{0.6, 0.7, 0.8, 0.9\}$, individual-clip $R^2 = 1.000$) — two separate architectures landing on the same specific wrong-but-confident behavior on the same synthetic system. Two of DynamiCrafter’s five remaining systems (pendulum angular frequency, spring-mass) are additionally bounded by a hard, non-adjustable model limitation: DynamiCrafter emits a fixed 16 frames per clip, too few for even one oscillation period at these systems’ lower grid values regardless of any frame-rate setting, so those two systems cannot be tested on this model at all (confirmed empirically at 96.7–100% dropped, not merely predicted).

Model	System	fit-quality dropped	slope β (in), 95% CI
LTX-Video	pendulum ω / ζ	100% / 100%	n/a
	bouncing ball / spring-mass	94.3% / 100%	n/a
	inclined slide	68.6%	−0.464 [−1.751, 1.097]
CogVideoX	pendulum ω / ζ	100% / 100%	n/a
	bouncing ball (all seeds $\to \hat{e} \approx 1$)	97.1%	n/a
	spring-mass / inclined slide	100% / 100%	n/a
DynamiCrafter	pendulum ω / ζ[a]	100% / 96.7%	n/a
	bouncing ball (seed-4 $\to \hat{e} \approx 1$)	74.3%	n/a
	spring-mass[a]	100%	n/a
	inclined slide	40.0%	0.521 [−6.185, 6.850]

Table 3: Generality across systems (E4), all under C2. “n/a” means too few clips survived the fit-quality gate for a slope estimate; this is itself the finding for those rows (Section 4.4). [a]Bounded by DynamiCrafter’s fixed 16-frame output, not just conditioning failure — see text.

A third mechanism variant: a split, not a single, default. DynamiCrafter’s inclined-slide result contains a pattern the two-hypothesis framework of Section 3 does not name. One seed behaves like the single-default cases above: $\hat{\mu} \approx 1.80$ across the whole grid, in-range and out-of-range together. A second seed instead shows *two* distinct, individually tight clusters — $\hat{\mu} \approx 4.7$ (std $= 0.04$) for every in-range request, and a different $\hat{\mu} \approx 7.6$ (std $= 0.13$) for every out-of-range request — rather than one value spanning both. This is neither “ignores θ entirely” (the value does shift) nor a single global default (there are two regimes, not one): it looks like the model has learned a coarse in-range versus out-of-range distinction without learning anything continuous inside either regime. We name this the *split-default* pattern and flag it as an open case for the H1/H2 framework, not yet folded into PRI or CBI.

E7, mechanism adjudication. Applying `select_mechanism` (H1 shrinkage vs. H2 clamp, Section 4) to CogVideoX's out-of-range projectile data (12 gated points spanning $g \in \{1.6, 3, 20\}$) selects *neither* by the function's leave-one-out gate, but not because the data are unstructured. With θ_0 estimated jointly, PRI $= 0.97$ (near-total insensitivity to the requested g) and the estimated default $\hat{\theta}_0 = -10.56$ lands almost exactly on the seed-cluster values already observed in-range; AIC prefers the H1 shrinkage shape over the H2 clamp shape by a wide margin (ΔAIC $= -47.9$). What fails is the leave-one-θ-out R^2 (-0.50), and the likely reason is visible in the raw data: H1 as specified assumes *one* global θ_0, but the actual reversion target is seed-specific (Section 4.3). Pooling several seeds' distinct fixed points and fitting a single θ_0 necessarily leaves structured, not random, residual error, which is exactly what defeats a leave-one-out test even when the qualitative shape ("reverts to something roughly fixed") is correct. The paired-bootstrap extrapolation test is unambiguous on its own: $\mathrm{PRE}(S_{\mathrm{out}}) - \mathrm{PRE}(S_{\mathrm{in}}) = 1.068$, 95% CI $[0.267, 1.891]$, one-sided $p = 0.0055$. LTX-Video could not be adjudicated at all: only three of fifteen out-of-range clips survived the fit-quality gate, below the minimum of four `select_mechanism` requires.

Reading the outcomes honestly. None of the four outcomes anticipated in the protocol is a clean fit, and we report the actual shape of the result rather than forcing one. This is closest to outcome (2), *no conditioning effect*: under C2, on the in-range grid, all three models' faithfulness slope is statistically indistinguishable from zero (Table 2), so "open generators do not honor conditioned dynamics" is the correct headline for the scope actually tested. But the finding is more specific than a flat null. Two of the three models do not fail by ignoring the request in an unstructured way; they confidently generate physically coherent motion at one of a small number of *fixed, wrong* values, selected by the generation seed rather than the requested θ, and this specific signature reproduces across two independent model families and at least four distinct physics systems (projectile, spring-mass, bouncing-ball, inclined-slide). That is structurally close to outcome (4), *neither H1 nor H2 fits cleanly*, but for an identifiable and informative reason — the paper's H1 test assumes a single global default, and the data show a *seed-conditional* default instead, a distinction the current formalism does not have a parameter for. The inclined-slide split-default pattern (Section 4.5) is a further, third variant along the same axis. We treat naming this family of mechanisms — confident convergence to a small set of seed- or regime-selected fixed points, rather than either smooth shrinkage to one default or a clamp at the range edge — as the paper's empirical contribution in place of the originally hypothesized clean H1/H2 split, and we do not force the PRI/CBI labels where the leave-one-out test says they do not fit.

9 Limitations

The suite is intentionally synthetic and 2D, which strengthens identifiability but limits ecological validity; conclusions are about controllable low-dimensional dynamics, not arbitrary natural scenes. The default-prior estimate θ_0 is model-behavioral, not a claim about training data we cannot observe.

Several further limitations are specific to what was actually run rather than what was planned, and we state them plainly. **Channel.** Every result is under C2 (text-specified) conditioning; C1 (frame-implied, the protocol's primary channel) was not tested end-to-end for any model, because the diffusers component needed for genuine multi-frame conditioning on LTX-Video is broken in the release used here, and CogVideoX's and DynamiCrafter's official I2V pipelines accept only a single conditioning image regardless. It is possible, and untested, that frame-implied conditioning would elicit different behavior than naming the parameter in text. **Coverage.** E7's mechanism adjudication was only possible on one model (CogVideoX) and one system (projectile): LTX-Video's low fit-quality yield left too few out-of-range points to adjudicate at all, and DynamiCrafter's most novel result (the split-default pattern on inclined-slide, Section 8) has not been run through `select_mechanism`. The seed-conditional-default finding itself is therefore a robust, cross-model, cross-system *observation*, but a formal mechanism adjudication of it exists for only one model-system pair. **Model ceiling effects.** DynamiCrafter's fixed 16-frame output makes two of six systems (pendulum angular frequency, spring-mass) untestable on that model for a reason unrelated to conditioning faithfulness, which we distinguish explicitly from the conditioning failures elsewhere (Section 8) but which nonetheless reduces the number of model-system pairs with usable data. **Plausibility contrast (E3) and the controllability audit (E9)** were not run in this draft; the central faithfulness result (E0/E1/E4/E7) stands independently of them, but the paper's Contribution 1 (the plausibility-versus-faithfulness gap) and Contribution 4 (auditing controllability claims) are correspondingly not yet empirically supported. None of these bound the core measurement's validity (Section 8, E6), but each bounds the scope of what the results license us to claim.

10 Conclusion

We reframe "learning physics from video" for a generator as a falsifiable, label-free question of parameter faithfulness and extrapolation, and provide a black-box diagnostic, metrics with a stated identifiability condition, a controllable suite, and a protocol whose headline test separates plausibility from correctness. Applied to three frozen, off-the-shelf I2V generators, the diagnostic finds that none detectably honors a text-specified physical parameter even within the everyday range, contrary to the field's implicit premise of at least local faithfulness. The failure is not simply an absence of signal: two of the three models confidently generate physically coherent motion that converges to a small number of fixed, wrong values selected by the generation seed rather than the request, a specific and reproducible mechanism that appears across independent model families and multiple physics systems, and that the field's existing two-hypothesis account of out-of-range failure does not have a clean category for. The artifact — suite, pipeline, and metrics — is reusable on any image-to-video model, including future ones that claim physics controllability, and the mechanism identified here is a concrete target for follow-up work: whether it reflects prior reversion at the level of the sampling seed rather than the model as a whole, and whether it persists under frame-implied (C1) conditioning, which this draft was unable to test end-to-end (Section 8).

Reproducibility statement. All systems are procedurally generated from a seeded deterministic engine; models are public; trackers and fitting are off-the-shelf or closed-form. The analysis code (metrics, bootstrap CIs, synthetic smoke test) and the system generators are released. No human-annotated labels are used. We will fill the ICLR reproducibility checklist at submission per the current author guide.

Ethics statement. The work uses synthetic scenes and public models, with no human-subjects data beyond a small voluntary plausibility-rating subset (to be collected under the venue's guidelines). The diagnostic is intended to prevent overclaiming about video generators as physical simulators, which has safety relevance in robotics and scientific settings.

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A Failure-form index estimation

H1 (shrinkage, PRI**).** Given pairs $(\theta_i, \hat{\theta}_{ij})$, fit $\hat{\theta} = \alpha\theta + (1-\alpha)\theta_0$ by least squares with θ_0 either fixed to the physically typical value or estimated jointly; $\mathrm{PRI} = 1 - \hat{\alpha}$, clipped to $[0,1]$. **H2 (clamp,** CBI**).** Fit $\hat{\theta} = \min(\theta, \theta_{\text{edge}})$ with θ_{edge} at the in-range boundary (or estimated); $\mathrm{CBI} = 1 - \mathrm{RSS}_{\mathrm{H2}}/\mathrm{RSS}_{\mathrm{null}}$. **Selection.** Compare H1 and H2 by leave-one-θ-out R^2 and by ΔAIC; if both fits are poor, report the observed behavior without forcing a label. All confidence intervals use the paired bootstrap over (θ_i, seed).

B Consistency of the recovery estimator: full statement and proof

We restate Proposition 1 with explicit regularity conditions and give the argument in full. The claim concerns only the estimator R, not the generator G (see the Remark in Section 4).

Setup. Fix a system with law f and scalar parameter $\theta \in \Theta$, with Θ a compact interval. A clip of T frames at resolution scale yields a tracked observable $o \in \mathbb{R}^d$ (object centers, angles, or contact times). Write the population fitting statistic as $T(\theta) \in \mathbb{R}^k$ (for the projectile, the quadratic coefficient of height versus time; for the pendulum, the mean zero-crossing interval; for the bouncing ball, the successive height ratio). The estimator forms an empirical statistic $\hat{T}$ from o and returns $\hat{\theta} = g(\hat{T})$, where $g = T^{-1}$ on the range of T.

Assumptions. (A1, trackability) the tracker returns $\hat{o} = o^\star + \varepsilon$ with $\mathbb{E}[\varepsilon] = 0$ and $\mathrm{Var}(\varepsilon) = \sigma^2\Sigma$ for a fixed positive-definite Σ, and $\sigma^2 \to 0$ as T and the resolution grow; (A2, identifiability) $T : \Theta \to T(\Theta)$ is injective and continuously differentiable with $\|T'(\theta)\| \geq c > 0$ on Θ; (A3, regularity) the empirical statistic is a continuous functional $\hat{T} = \Phi(\hat{o})$ with Φ continuous at $o^\star$ and $\Phi(o^\star) = T(\theta)$.

Proposition 2 (Consistency, restated)**.** *Under (A1)–(A3),* $\hat{\theta} = g(\Phi(\hat{o})) \xrightarrow{p} \theta$ *as* $\sigma^2 \to 0$.

Proof. By (A2), T is a continuous injection on the compact set Θ, so its inverse $g = T^{-1}$ is continuous on $T(\Theta)$ (a continuous bijection from a compact space to a Hausdorff space is a homeomorphism). By (A1), $\hat{o} \xrightarrow{p} o^\star$ since $\mathrm{Var}(\hat{o}) = \sigma^2\Sigma \to 0$ and the mean is $o^\star$ (convergence in mean square implies convergence in probability). By (A3), Φ

is continuous at $o^\star$, so the continuous mapping theorem gives $\hat{T} = \Phi(\hat{o}) \xrightarrow{p} \Phi(o^\star) = T(\theta)$. Applying the continuous mapping theorem again with the continuous g yields $\hat{\theta} = g(\hat{T}) \xrightarrow{p} g(T(\theta)) = \theta$. $\square$

Per-system check of (A2)–(A3). *Projectile:* $T(\theta) = -\theta/2$ is linear (hence injective, $T' = -1/2$) and Φ is ordinary least squares on a quadratic basis, continuous in the data. *Pendulum (small angle):* $T(\theta) = 2\pi/\theta$ on $\theta > 0$ is strictly monotone with nonvanishing derivative on any Θ bounded away from 0; zero-crossing interpolation is continuous where crossings are simple. *Bouncing ball:* the map $e \mapsto e^2$ from restitution to successive height ratio is injective and smooth on $(0,1]$. The fit-quality gate ($R^2 < \rho$) excludes generations on which Φ is ill-posed (object morphs or vanishes), so (A3) is enforced operationally rather than assumed. *Independent human verification of this appendix is still required before submission.*

C System grids and rendering details

All scenes are rendered by a seeded deterministic 2D engine at 256×256 over $T{=}24$ frames at 24 fps, with a single high-contrast disk (radius 8 px) on a plain background and a fixed ground line; only the swept parameter and the seeded initial condition vary within a system. Spatial and temporal scales are held fixed within a clip, and all metrics use the ratio $\hat{\theta}/\theta$ so the result is invariant to a common rescaling (Section 4). The in-range split is centered on values common in natural video; the out-of-range split probes rare or absent values. Example grids (final grids are fixed in the released config and reported with results):

System	θ	**In-range grid**	**Out-of-range grid**
Projectile	g	$\{5, 7, 9.8, 12, 15\}$	$\{1.6, 3, 20, 25\}$
Damped pendulum	ω	$\{2, 3, 4, 5\}$ rad/s	$\{0.5, 1, 8, 12\}$
	ζ	$\{0.02, 0.05, 0.1\}$	$\{0.3, 0.6, 0.9\}$
Bouncing ball	e	$\{0.6, 0.7, 0.8, 0.9\}$	$\{0.2, 0.35, 0.97\}$
Spring–mass	k	$\{20, 40, 60, 80\}$ N/m	$\{5, 10, 200, 400\}$
Inclined slide	μ	$\{0.1, 0.2, 0.3, 0.4\}$	$\{0.01, 0.7, 1.0\}$

Table 4: Example sweep grids per system (illustrative; the released config fixes the final values). Each row has one swept axis; $m \geq 5$ seeded initial conditions per grid point.